Types of Arguments in Python

- Types of arguments:
  - Positional arguments
  - Keyword arguments
  - Default arguments
  - Variable-length arguments (*args, **kwargs)
- Functions can return multiple values.

explain each with program in python

Sure! Let's break it down **clearly** with explanations and Python examples for each type of argument, and also show how functions can return multiple values. I'll go **step by step**.

1️⃣ Positional Arguments

Explanation:

These are the most common type. The arguments are passed to the function **in order**, and the position matters.

Example:

Python

```python
def add(a, b):
    return a + b

result = add(5, 3)  # 5 → a, 3 → b
print("Sum:", result)
```

Output:

Code

```
Sum: 8
```

✅ Here, `5` is assigned to `a` and `3` to `b` because of their positions.

2️⃣ Keyword Arguments

Explanation:

Arguments are passed with their **parameter names**, so the order doesn't matter.

Example:

```python
def introduce(name, age):
    print(f"My name is {name} and I am {age} years old.")

introduce(age=25, name="Chetan")
```

Output:

```
My name is Chetan and I am 25 years old.
```

✅ Using `name=` and `age=` allows us to pass arguments in any order.

3️⃣ Default Arguments

Explanation:

You can assign a **default value** to a parameter. If no value is provided during the function call, the default is used.

Example:

```python
def greet(name, msg="Hello"):
    print(f"{msg}, {name}!")

greet("Chetan")         # Uses default msg
greet("Chetan", "Hi")    # Overrides default msg
```

Output:

```
Hello, Chetan!
Hi, Chetan!
```

✅ `msg` has a default value `"Hello"` but can be overridden.

4️⃣ Variable-length Arguments (`*args` and `**kwargs`)

a) `*args` (Non-keyword variable arguments)

Explanation:

Allows a function to accept **any number of positional arguments**.

Example:

```python
def sum_all(*numbers):
    total = 0
    for num in numbers:
        total += num
    return total

print(sum_all(1, 2, 3, 4, 5))
```

Output:

```
15
```

✅ `*numbers` collects all positional arguments as a tuple.

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b) `**kwargs` (Keyword variable arguments)

Explanation:

Allows a function to accept **any number of keyword arguments**.

Example:

```python
def show_info(**info):
    for key, value in info.items():
        print(f"{key}: {value}")

show_info(name="Chetan", age=21, city="Wardha")
```

Output:

```
```

```
name: Chetan
age: 21
city: Wardha
```

✅ `**info` collects all keyword arguments as a dictionary.

5️⃣ Functions Returning Multiple Values

Explanation:

Python functions can return **multiple values** using tuples.

Example:

```python
def arithmetic_operations(a, b):
    sum_ = a + b
    diff = a - b
    prod = a * b
    return sum_, diff, prod  # Returns 3 values as a tuple

s, d, p = arithmetic_operations(10, 5)
print("Sum:", s, "Difference:", d, "Product:", p)
```

Output:

```
Sum: 15 Difference: 5 Product: 50
```

✅ Multiple values are returned as a tuple and can be unpacked into separate variables.

If you want, I can make **one single Python program combining all these argument types and multiple returns** so you can see them all together in action.

Do you want me to do that?